



Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Semester: (Fall, Year: 2024), B.Sc. in CSE (Day)

Cpu Scheduler

Course Title: Operating System Lab
Course Code: CSE-310
Section: 222-D6

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<u>Lab Project Status</u>	
Marks:	Signature:
Comments:	Date:

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Chapter 1

Introduction

1.1 Overview

The CPU Scheduler is a simulation program designed to replicate the operation of various CPU scheduling algorithms used in operating systems. CPU scheduling is a fundamental concept in process management, ensuring that processes are executed efficiently while optimizing system performance. This project implements and analyzes five widely used scheduling algorithms:

- **First-Come-First-Serve (FCFS):** Processes are scheduled in the order of their arrival.
- **Shortest Job First (SJF):** Processes with the shortest burst time are prioritized.
- **Shortest Remaining Time First (SRTF):** A preemptive version of SJF that dynamically selects processes with the shortest remaining time.
- **Round Robin (RR):** Processes are scheduled cyclically using a predefined time quantum.
- **Priority Scheduling:** Processes are scheduled based on their assigned priorities.

This program is designed to calculate essential metrics, such as waiting time, turnaround time, and completion time, for each algorithm. It provides a user-friendly interface and logs detailed execution data, making it a valuable tool for students, researchers, and educators.

1.2 Motivation

Efficient CPU scheduling is critical for ensuring optimal utilization of CPU resources and minimizing process waiting times in multi-tasking environments. The need for a comprehensive and interactive tool that bridges theoretical knowledge with practical application inspired the development of this project. Existing tools often lack user interactivity or fail to provide detailed analysis of scheduling metrics.

Key motivations include:

- Enhancing understanding of CPU scheduling concepts through visualization.
- Providing a practical platform to compare scheduling algorithms in real-time.
- Supporting educators and students in exploring complex scheduling scenarios with varying arrival times, burst times, and priorities.
- Facilitating research by offering detailed performance metrics and execution logs.

1.3 Problem Definition

1.3.1 Problem Statement

The challenge addressed by the CPU Scheduler is the absence of an interactive, modular, and detailed tool to simulate CPU scheduling algorithms. Many existing solutions are either overly simplified or fail to provide insights into critical scheduling metrics. This project aims to fill this gap by offering a simulator that is not only user-friendly but also versatile enough to handle different scheduling scenarios and provide detailed performance analysis.

1.3.2 Complex Engineering Problem

The CPU Scheduler tackles a complex engineering problem involving the interplay of multiple algorithms, varying process attributes, and system-level constraints. The attributes of complexity addressed by this project include:

Attribute	Explanation
Depth of knowledge required	Involves understanding of process management, algorithm design, and implementation in Bash scripting.
Range of conflicting requirements	Balances usability, modularity, and computational efficiency while providing detailed insights into scheduling metrics.
Depth of analysis required	Requires comparative analysis of five distinct scheduling algorithms under varying input conditions.
Familiarity of issues	Simulates real-world CPU scheduling challenges such as handling priorities, time quanta, and preemption.
Extent of stakeholder involvement	Addresses the needs of students, educators, and researchers through its interactive interface and detailed logging capabilities.

Table 1.1: Attributes of Complex Engineering Problems

1.4 Design Goals/Objectives

The CPU Scheduler is designed with the following objectives:

- **Simulation of Algorithms:** To implement and simulate the operation of five key CPU scheduling algorithms.
- **Comprehensive Analysis:** To compute and display critical scheduling metrics, including waiting time, turnaround time, and average metrics.
- **User-Friendly Interface:** To provide an interactive menu-driven interface for easy algorithm selection and input handling.
- **Modularity:** To maintain a modular structure, with separate scripts for each algorithm, ensuring maintainability and scalability.
- **Detailed Logging:** To generate execution logs for performance analysis and debugging.
- **Educational Value:** To serve as an educational tool for teaching and learning CPU scheduling concepts effectively.

1.5 Application

The CPU Scheduler has significant real-world and academic applications, including:

- **Educational Tool:**
 - Helps students grasp CPU scheduling concepts by visualizing algorithm execution.
 - Provides instructors with a practical tool to demonstrate scheduling techniques.
- **Research and Analysis:**
 - Supports researchers in evaluating and comparing scheduling algorithms under different conditions.
 - Offers detailed logs for in-depth performance analysis.
- **System Design:**
 - Assists system architects in choosing appropriate scheduling algorithms for specific environments.
 - Enables testing of system performance under various scheduling scenarios.
- **Debugging and Testing:**
 - Serves as a platform for testing the impact of scheduling strategies on process execution.

- Identifies bottlenecks and inefficiencies in scheduling algorithms.
- **Process Optimization:**
 - Provides insights into optimizing scheduling parameters like time quantum and priority levels for better system performance.

Chapter 2

Design/Development/Implementation

2.1 Introduction

The **CPU Scheduler** project aims to provide an interactive and modular simulation of widely used CPU scheduling algorithms. It integrates theoretical concepts with practical implementation to analyze key scheduling metrics such as waiting time, turnaround time, and completion time. The system is designed to support five algorithms: First-Come-First-Serve (FCFS), Shortest Job First (SJF), Shortest Remaining Time First (SRTF), Round Robin (RR), and Priority Scheduling. This chapter elaborates on the architectural design, workflow, tools, and algorithmic implementation of the project.

2.2 Project Details

2.2.1 Overall Architecture

The project follows a modular structure, enabling ease of development and scalability. The components include:

- **Main Script** (`main.sh`): The central script that provides a menu-driven interface for user interaction. It connects all the algorithm-specific scripts.
- **Algorithm Scripts**: Separate scripts for each scheduling algorithm:
 - `fcfs.sh`: Implements the First-Come-First-Serve scheduling algorithm.
 - `sjf.sh`: Implements the Shortest Job First scheduling algorithm.
 - `srtf.sh`: Implements the Shortest Remaining Time First scheduling algorithm.
 - `rr.sh`: Implements the Round Robin scheduling algorithm.
 - `priority.sh`: Implements the Priority Scheduling algorithm with preemptive and non-preemptive modes.
- **Configuration File**: Stores customizable parameters such as time quantum, logging preferences, and color settings.

2.2.2 Subsystem Integration

The `main.sh` script acts as the controller, invoking the appropriate scheduling algorithm script based on the user's choice. This design ensures modularity, allowing each algorithm to function independently while being seamlessly integrated into the main program.

2.3 Implementation

2.3.1 Workflow

The implementation workflow involves:

1. **User Interaction:** The user interacts with the `main.sh` script, which displays a menu for selecting a scheduling algorithm.
2. **Input Collection:** Each algorithm script prompts the user for process details, including arrival time, burst time, and priority (if applicable). Inputs are validated for correctness.
3. **Algorithm Execution:** The selected algorithm schedules processes and calculates metrics like waiting time, turnaround time, and completion time.
4. **Results Display:** Metrics are presented in a tabular format, with color-coded outputs for readability.
5. **Logging:** Execution details are logged, as configured in the settings, for further analysis.

2.3.2 Tools and Libraries

- **Bash Scripting:** The entire project is implemented in Bash, utilizing its capabilities for command-line interaction and process control.
- **awk and bc:** Used for arithmetic calculations and formatted output.
- **Log Files:** Results and execution details are stored in log files for debugging and performance evaluation.

2.3.3 Implementation Details

Each algorithm script is implemented as follows:

First-Come-First-Serve (FCFS)

The `fcfs.sh` script schedules processes in the order of their arrival. Completion times are calculated sequentially, with each process starting immediately after the previous one finishes.

Shortest Job First (SJF)

The `sjf.sh` script selects the process with the shortest burst time. Processes are sorted by burst time, and metrics are calculated accordingly.

Shortest Remaining Time First (SRTF)

The `srtf.sh` script implements a preemptive version of SJF. It dynamically selects the process with the shortest remaining burst time at each time unit.

Round Robin (RR)

The `rr.sh` script schedules processes cyclically, using a user-specified time quantum. Processes exceeding the time quantum are returned to the queue.

Priority Scheduling

The `priority.sh` script allows users to choose between preemptive and non-preemptive modes. Processes are scheduled based on priority, with ties resolved by arrival time.

2.4 Algorithms

The algorithms implemented in this project are described below:

2.4.1 First-Come-First-Serve (FCFS)

Data: Processes with Arrival Times and Burst Times

Result: Completion Time, Waiting Time, Turnaround Time

```
1 Sort processes by Arrival Time;
2 foreach Process do
3   if Current Time < Arrival Time then
4     | Wait until the process arrives;
5   else
6     | Execute the process;
7   end
8   | Update Completion Time;
9 end
10 Calculate Waiting Time and Turnaround Time for each process;
```

2.4.2 Shortest Job First (SJF)

Data: Processes with Arrival Times and Burst Times

Result: Completion Time, Waiting Time, Turnaround Time

```
1 Sort processes by Burst Time;
2 foreach Process do
3   |   Select the process with the smallest burst time;
4   |   Execute and update Completion Time;
5 end
6 Calculate Waiting Time and Turnaround Time for each process;
```

2.4.3 Shortest Remaining Time First (SRTF)

Data: Processes with Arrival Times and Burst Times

Result: Completion Time, Waiting Time, Turnaround Time

```
1 while All processes are not completed do
2   |   Select the process with the shortest remaining burst time;
3   |   Execute for one time unit;
4   |   Update remaining time and Completion Time;
5 end
6 Calculate Waiting Time and Turnaround Time for each process;
```

2.4.4 Round Robin (RR)

Data: Processes with Arrival Times, Burst Times, and Time Quantum

Result: Completion Time, Waiting Time, Turnaround Time

```
1 while All processes are not completed do
2   |   foreach Process in the queue do
3   |   |   Execute for Time Quantum or remaining Burst Time;
4   |   |   Update remaining time and Completion Time;
5   |   |   Return unfinished processes to the queue;
6   |   end
7 end
8 Calculate Waiting Time and Turnaround Time for each process;
```

2.4.5 Priority Scheduling

Data: Processes with Arrival Times, Burst Times, and Priorities

Result: Completion Time, Waiting Time, Turnaround Time

- 1 **if** *Preemptive Mode* **then**
 - 2 Select the highest-priority process dynamically;
 - 3 Execute until preempted or completed;
 - 4 **end**
 - 5 Select the highest-priority process;
 - 6 Execute until completion;
 - 7 Calculate Waiting Time and Turnaround Time for each process;
-

2.5 Coding

2.5.1 Main Script: CPU Scheduler

Below is the Bash script for the main program of the CPU Scheduling Simulator:

```
1  #!/bin/bash
2
3  # Define color codes for text formatting
4  NC='\033[0m'           # No Color
5  RED='\033[0;31m'       # Red
6  GREEN='\033[0;32m'     # Green
7  YELLOW='\033[0;33m'    # Yellow
8  BLUE='\033[0;34m'      # Blue
9  CYAN='\033[0;36m'      # Cyan
10 MAGENTA='\033[0;35m'   # Magenta
11 BOLD='\033[1m'         # Bold
12 UNDERLINE='\033[4m'    # Underline
13 BLINK='\033[5m'        # Blink
14 BG_BLUE='\033[44m'     # Blue background
15 BG_YELLOW='\033[43m'   # Yellow background
16
17 # Function to display the header
18 display_header() {
19     echo -e "${BG_BLUE}${BOLD}${CYAN}
20     }#####
21     ${NC}"
22     echo -e "${BG_BLUE}${BOLD}${CYAN}# CPU Scheduling
23     Simulation Program          ${NC}"
24     echo -e "${BG_BLUE}${BOLD}${CYAN}
25     }#####
26     ${NC}"
27     echo -e "${NC}"
28 }
29
30 # Function to display the main menu
31 show_menu() {
32     echo -e "${YELLOW}${BOLD}Please select a scheduling algorithm to
33     run:${NC}"
34     echo -e ""
```

```

29 echo -e "${BLUE}[1]${NC} ${GREEN}${BOLD}First-Come-First-Serve (
    FCFS)${NC}"
30 echo -e "${BLUE}[2]${NC} ${GREEN}${BOLD}Shortest Job First (SJF)
    ${NC}"
31 echo -e "${BLUE}[3]${NC} ${GREEN}${BOLD}Shortest Remaining Time
    First (SRTF)${NC}"
32 echo -e "${BLUE}[4]${NC} ${GREEN}${BOLD}Round Robin (RR)${NC}"
33 echo -e "${BLUE}[5]${NC} ${GREEN}${BOLD}Priority Scheduling (
    Preemptive/Non-Preemptive)${NC}"
34 echo -e "${RED}[Q]${NC} ${BOLD}Exit${NC}"
35 echo -e ""
36 echo -e "${CYAN}Enter your choice: ${NC}"
37 }
38
39 # Function to process the user input and run the respective
    algorithm
40 process_choice() {
41     case $1 in
42         1)
43             echo -e "${GREEN}${BOLD}Running First-Come-First-Serve (FCFS
                )...${NC}"
44             ./fcfs.sh                # Execute FCFS Scheduling Algorithm
45             ;;
46         2)
47             echo -e "${GREEN}${BOLD}Running Shortest Job First (SJF)...$
                {NC}"
48             ./sjf.sh                # Execute SJF Scheduling Algorithm
49             ;;
50         3)
51             echo -e "${GREEN}${BOLD}Running Shortest Remaining Time
                First (SRTF)...${NC}"
52             ./srtf.sh                # Execute SRTF Scheduling Algorithm
53             ;;
54         4)
55             echo -e "${GREEN}${BOLD}Running Round Robin (RR)...${NC}"
56             ./rr.sh                # Execute Round Robin Scheduling Algorithm
57             ;;
58         5)
59             echo -e "${GREEN}${BOLD}Running Priority Scheduling...${NC}"
60             ./priority.sh            # Execute Priority Scheduling Algorithm
61             ;;
62         [Qq])
63             echo -e "${RED}${BOLD}Exiting program...${NC}"
64             exit 0                    # Exit the program
65             ;;
66         *)
67             echo -e "${RED}${BOLD}Invalid option! Please choose a valid
                option.${NC}"
68             ;;
69     esac
70 }
71
72 # Function to display an exit message
73 display_exit_message() {
74     echo -e "${BG_YELLOW}${BOLD}${MAGENTA}Thank you for using the
        CPU Scheduling Simulator!${NC}"
75     echo -e "${BG_YELLOW}${BOLD}${MAGENTA}Have a great day ahead!${
        NC}"

```

```

76 }
77
78 # Main script execution
79 while true; do
80     display_header          # Display the decorated header
81     show_menu              # Show the menu of available
82                             algorithms
83     read choice            # Read user choice
84     process_choice "$choice" # Execute the selected scheduling
85                             algorithm
86     echo -e "\n${CYAN}Press any key to return to the main menu...${
87         NC}"
88     read -n 1 -s          # Wait for user input to continue
89 done
90
91 display_exit_message      # Display exit message when the
92                             program ends

```

Listing 2.1: Main Script for CPU Scheduling Simulation Program

2.5.2 First-Come-First-Serve (FCFS) Scheduling Script

Below is the Bash script implementing the FCFS scheduling algorithm:

```

1  #!/bin/bash
2
3  # Global Arrays
4  declare -A arrival_time
5  declare -A burst_time
6  declare -A completion_time
7  declare -A waiting_time
8  declare -A turn_around_time
9
10 # Function: Display Header
11 display_fcfs_header() {
12     echo -e "\n##### FCFS Scheduling
13     #####"
14     echo -e "Executes processes in the order of their arrival time."
15 }
16
17 # Function: Collect Process Data
18 collect_process_data() {
19     echo -e "\nEnter the number of processes:"
20     read num_processes
21
22     while [[ ! "$num_processes" =~ ^[0-9]+$ || $num_processes -le 0
23         ]]; do
24         echo -e "Invalid input! Please enter a positive integer."
25         read num_processes
26     done
27
28     for ((i=1; i<=num_processes; i++)); do
29         echo -e "Enter Arrival Time for Process $i:"
30         read arrival_time[$i]
31         while [[ ! "${arrival_time[$i]}" =~ ^[0-9]+$ ]]; do

```

```

30     echo -e "Invalid input! Enter a non-negative integer for
      arrival time."
31     read arrival_time[$i]
32 done
33
34     echo -e "Enter Burst Time for Process $i:"
35     read burst_time[$i]
36     while [[ ! "${burst_time[$i]}" =~ ^[0-9]+$ || "${burst_time[$i]}"
      -le 0 ]]; do
37         echo -e "Invalid input! Enter a positive integer for burst
      time."
38         read burst_time[$i]
39     done
40 done
41 }
42
43 # Function: Calculate Completion Time
44 calculate_completion_time() {
45     completion_time[1]=$((arrival_time[1] + burst_time[1]))
46     for ((i=2; i<=num_processes; i++)); do
47         if ((arrival_time[i] > completion_time[i-1])); then
48             completion_time[i]=$((arrival_time[i] + burst_time[i]))
49         else
50             completion_time[i]=$((completion_time[i-1] + burst_time[i]))
51         fi
52     done
53 }
54
55 # Function: Calculate Waiting Time
56 calculate_waiting_time() {
57     for ((i=1; i<=num_processes; i++)); do
58         waiting_time[$i]=$((completion_time[$i] - arrival_time[$i] -
      burst_time[$i]))
59     done
60 }
61
62 # Function: Calculate Turnaround Time
63 calculate_turn_around_time() {
64     for ((i=1; i<=num_processes; i++)); do
65         turn_around_time[$i]=$((completion_time[$i] - arrival_time[$i]
      ))
66     done
67 }
68
69 # Function: Display Results
70 display_fcfs_results() {
71     echo -e "\nResults:"
72     echo -e "-----"
73     echo -e "Process ID\tArrival Time\tBurst Time\tCompletion Time\t
      tWaiting Time\tTurnaround Time"
74
75     total_waiting_time=0
76     total_turnaround_time=0
77
78     for ((i=1; i<=num_processes; i++)); do
79         echo -e "P$i\t\t${arrival_time[$i]}\t\t${burst_time[$i]}\t\t${
      completion_time[$i]}\t\t${waiting_time[$i]}\t\t${
      turn_around_time[$i]}"

```

```

80     total_waiting_time=$((total_waiting_time + waiting_time[$i]))
81     total_turnaround_time=$((total_turnaround_time +
      turn_around_time[$i]))
82 done
83
84 avg_waiting_time=$(awk "BEGIN {printf \"%.2f\\",
      $total_waiting_time / $num_processes}")
85 avg_turnaround_time=$(awk "BEGIN {printf \"%.2f\\",
      $total_turnaround_time / $num_processes}")
86
87 echo -e "-----"
88 echo -e "Average Waiting Time: $avg_waiting_time"
89 echo -e "Average Turnaround Time: $avg_turnaround_time"
90 }
91
92 # Main Function
93 main_fcfs() {
94     display_fcfs_header
95     collect_process_data
96     calculate_completion_time
97     calculate_waiting_time
98     calculate_turn_around_time
99     display_fcfs_results
100 }
101
102 # Run
103 main_fcfs

```

Listing 2.2: FCFS Scheduling Script

2.5.3 Shortest Job First (SJF) Scheduling Algorithm

Below is the Bash script implementing the Shortest Job First (SJF) scheduling algorithm:

```

1  #!/bin/bash
2
3  # Shortest Job First (SJF) Scheduling Algorithm with User Input
4
5  # Function to print decorated header
6  function print_header() {
7      clear
8      echo -e "\033[1;36m
      =====
9      echo -e "          \033[1;33mShortest Job First (SJF) Scheduling
      Algorithm\033[0m"
10     echo -e "\033[1;36m
      =====
11 }
12
13 # Function to perform SJF scheduling
14 function sjf_scheduling() {
15     # Prompt the user for the number of processes
16     echo -e "\n\033[1;32mEnter the number of processes: \033[0m"
17     read num_processes

```



```

18
19 # Declare arrays to store process details
20 processes=()
21 arrival_times=()
22 burst_times=()
23 completion_times=()
24 waiting_times=()
25 turnaround_times=()
26
27 total_waiting_time=0
28 total_turnaround_time=0
29
30 # User input for each process (Process ID, Arrival Time, Burst
    Time)
31 for ((i = 0; i < $num_processes; i++)); do
32     echo -e "\n\033[1;32mEnter details for Process $((i+1))
        :\033[0m"
33     echo -e "Enter Arrival Time for Process $((i+1)): "
34     read arrival_time
35     echo -e "Enter Burst Time for Process $((i+1)): "
36     read burst_time
37
38     processes+=("P$((i+1))")
39     arrival_times+=($arrival_time)
40     burst_times+=($burst_time)
41 done
42
43 # Sorting processes by burst time (Shortest Burst Time First)
44 for ((i = 0; i < $num_processes; i++)); do
45     for ((j = $i+1; j < $num_processes; j++)); do
46         if (( ${burst_times[$i]} > ${burst_times[$j]} )); then
47             # Swap processes, arrival times, and burst times
48             temp="${processes[$i]}"
49             processes[$i]="${processes[$j]}"
50             processes[$j]="$temp"
51             temp="${arrival_times[$i]}"
52             arrival_times[$i]="${arrival_times[$j]}"
53             arrival_times[$j]="$temp"
54             temp="${burst_times[$i]}"
55             burst_times[$i]="${burst_times[$j]}"
56             burst_times[$j]="$temp"
57         fi
58     done
59 done
60
61 # Calculate completion, waiting, and turnaround times
62 completion_times[0]=${burst_times[0]}
63 waiting_times[0]=0
64 turnaround_times[0]=${completion_times[0]}
65
66 for ((i = 1; i < $num_processes; i++)); do
67     completion_times[$i]=$(( ${completion_times[$i-1]} + ${
        burst_times[$i]} ))
68     waiting_times[$i]=$(( ${completion_times[$i-1]} - ${
        arrival_times[$i]} ))
69     turnaround_times[$i]=$(( ${waiting_times[$i]} + ${
        burst_times[$i]} ))
70

```

```

71      # Cumulative totals
72      total_waiting_time=$((total_waiting_time + ${
73          waiting_times[$i]})
74      total_turnaround_time=$((total_turnaround_time + ${
75          turnaround_times[$i]})
76      done
77
78      # Display results in a table format
79      echo -e "\n\033[1;36mSJF Scheduling Results:\033[0m"
80      echo -e "\033[1;32m
81          -----\033[0m"
82      echo -e "Process ID      Arrival Time      Burst Time
83          Completion Time      Waiting Time      Turnaround Time"
84      echo -e "\033[1;32m
85          -----\033[0m"
86
87      for ((i = 0; i < $num_processes; i++)); do
88          echo -e "${processes[$i]} \t\t ${arrival_times[$i]} \t\t $
89              {burst_times[$i]} \t\t ${completion_times[$i]} \t\t ${
90                  waiting_times[$i]} \t\t ${turnaround_times[$i]}"
91      done
92      echo -e "\033[1;32m
93          -----\033[0m"
94
95      # Calculate and display averages
96      avg_waiting_time=$((total_waiting_time / $num_processes))
97      avg_turnaround_time=$((total_turnaround_time / $num_processes
98          ))
99
100      echo -e "\033[1;33mAverage Waiting Time: $avg_waiting_time
101          \033[0m"
102      echo -e "\033[1;33mAverage Turnaround Time:
103          $avg_turnaround_time\033[0m"
104      }
105
106      # Main program starts here
107      print_header
108      sjf_scheduling
109
110      # Log the result to a log file (optional)
111      echo -e "Process ID Arrival Time Burst Time Completion Time
112          Waiting Time Turnaround Time" > "logs/sjf.log"
113      for ((i = 0; i < ${#processes[@]}; i++)); do
114          echo -e "${processes[$i]} ${arrival_times[$i]} ${burst_times[
115              $i]} ${completion_times[$i]} ${waiting_times[$i]} ${
116                  turnaround_times[$i]}" >> "logs/sjf.log"
117      done
118      echo -e "Average Waiting Time: $avg_waiting_time" >> "logs/sjf.log
119          "
120      echo -e "Average Turnaround Time: $avg_turnaround_time" >> "logs/
121          sjf.log"

```

Listing 2.3: SJF Scheduling Script

2.5.4 Shortest Remaining Time First (SRTF) Scheduling Algorithm

Below is the Bash script implementing the Shortest Remaining Time First (SRTF) scheduling algorithm:

```
1  #!/bin/bash
2
3  # Function to calculate waiting time and turnaround time for each
  process
4  calculate_times() {
5      total_waiting_time=0
6      total_turnaround_time=0
7      echo "
8          ===== "
9      echo -e "\033[1;34mProcess ID      Arrival Time      Burst Time
10         Completion Time      Waiting Time      Turnaround Time\033[0m"
11      echo "
12         ===== "
13      for i in "${!arrival_times[@]}"; do
14          waiting_time=${completion_times[$i]}-${arrival_times[$i]}-${
15              burst_times[$i]}
16          turnaround_time=${completion_times[$i]}-${arrival_times[$i]}
17
18          total_waiting_time=$((total_waiting_time + waiting_time))
19          total_turnaround_time=$((total_turnaround_time +
20              turnaround_time))
21
22          # Output with color formatting for readability
23          echo -e "P\033[1;32m${i+1}\033[0m                ${arrival_times[
24              $i]}                ${burst_times[$i]}                ${
25              completion_times[$i]}                \033[1;33m$waiting_time
26              \033[0m                \033[1;33m$turnaround_time\033[0m"
27      done
28
29      average_waiting_time=$(echo "$total_waiting_time / ${#
30          arrival_times[@]}" | bc -l)
31      average_turnaround_time=$(echo "$total_turnaround_time / ${#
32          arrival_times[@]}" | bc -l)
33
34      echo "
35          ===== "
36      echo -e "\033[1;35mAverage Waiting Time: $average_waiting_time
37          \033[0m"
38      echo -e "\033[1;35mAverage Turnaround Time:
39          $average_turnaround_time\033[0m"
40      echo "
41          ===== "
42  }
43
44  # Read input from user
45  echo -e "\033[1;36mEnter the number of processes:\033[0m"
46  read n
47
48  # Declare arrays for arrival time, burst time, completion time,
49  and remaining time
50  declare -a arrival_times
```

```

37 declare -a burst_times
38 declare -a remaining_times
39 declare -a completion_times
40
41 # Read process details
42 for ((i=0; i<n; i++)); do
43     echo -e "\033[1;32mEnter details for Process $((i+1)):\033[0m"
44     echo -e "\033[1;36mEnter Arrival Time for Process $((i+1))
45         :\033[0m"
46     read arrival_times[$i]
47     echo -e "\033[1;36mEnter Burst Time for Process $((i+1)):\033[0m
48         "
49     read burst_times[$i]
50     remaining_times[$i]=${burst_times[$i]} # Initially, remaining
51         time equals burst time
52     completion_times[$i]=0
53 done
54
55 # Initializing variables
56 current_time=0
57 processes_completed=0
58
59 # Array to keep track of process completion status
60 is_completed=()
61 for ((i=0; i<n; i++)); do
62     is_completed[$i]=0 # Initially, no processes are completed
63 done
64
65 # SRTF algorithm logic
66 echo -e "\033[1;34mStarting SRTF Scheduling Algorithm...\033[0m"
67 while [ $processes_completed -lt $n ]; do
68     # Find the process with the shortest remaining time that has
69     # arrived
70     shortest_time_remaining=-1
71     process_to_execute=-1
72
73     for ((i=0; i<n; i++)); do
74         if [ ${arrival_times[$i]} -le $current_time ] && [ ${
75             is_completed[$i]} -eq 0 ]; then
76             if [ $shortest_time_remaining -eq -1 ] || [ ${
77                 remaining_times[$i]} -lt $shortest_time_remaining ]; then
78                 shortest_time_remaining=${remaining_times[$i]}
79                 process_to_execute=$i
80             fi
81         fi
82     done
83
84     # If a valid process is found, execute it
85     if [ $process_to_execute -ne -1 ]; then
86         # Execute for 1 unit of time
87         remaining_times[$process_to_execute]=$((remaining_times[
88             $process_to_execute]-1))
89         current_time=$((current_time+1))
90
91         # If the process has finished execution
92         if [ ${remaining_times[$process_to_execute]} -eq 0 ]; then
93             is_completed[$process_to_execute]=1
94             completion_times[$process_to_execute]=$current_time

```

```

88     processes_completed=$((processes_completed+1))
89     echo -e "\033[1;33mProcess P$((process_to_execute+1))
        completed at time $current_time.\033[0m"
90     fi
91 else
92     # No processes ready to execute, increase current time
93     current_time=$((current_time+1))
94     fi
95 done
96
97 # Calculate times and print results
98 calculate_times

```

Listing 2.4: SRTF Scheduling Script

2.5.5 Round Robin Scheduling Algorithm

Below is the Bash script implementing the Round Robin scheduling algorithm:

```

1  #!/bin/bash
2
3  # Function to calculate and display the final output
4  calculate_times() {
5      total_waiting_time=0
6      total_turnaround_time=0
7      echo -e "\n\033[1;36m
        =====\033[0m"
8      echo -e "\033[1;32m                                ROUND ROBIN
        SCHEDULING RESULTS\033[0m"
9      echo -e "\033[1;36m
        =====\033[0m"
10     "
11     echo -e "\033[1;34mProcess ID      Arrival Time      Burst Time
        Completion Time      Waiting Time      Turnaround Time\033[0m"
12     echo -e "\033[1;36m
        =====\033[0m"
13
14     for i in "${!arrival_times[@]"; do
15         waiting_time=$((completion_times[$i]-{arrival_times[$i]}-{
            burst_times[$i]}
16         turnaround_time=$((completion_times[$i]-{arrival_times[$i]}
17
18         total_waiting_time=$((total_waiting_time + waiting_time))
19         total_turnaround_time=$((total_turnaround_time +
            turnaround_time))
20
21         # Output with color formatting for readability
22         echo -e " \033[1;32mP$((i+1))\033[0m \033[1;33m${
            arrival_times[$i]}\033[0m \033[1;33m${
            burst_times[$i]}\033[0m \033[1;36m${completion_times
            [$i]}\033[0m \033[1;35m${waiting_time}\033[0m
            \033[1;35m${turnaround_time}\033[0m"
23     done
24
25     average_waiting_time=$((echo "$total_waiting_time / ${#
        arrival_times[@]}" | bc -l)

```

```

25 average_turnaround_time=$(echo "$total_turnaround_time / ${#
    arrival_times[@]}" | bc -l)
26
27 echo -e "\033[1;36m
    =====\033[0m
    "
28 echo -e "\033[1;35mAverage Waiting Time: \033[0
    m$average_waiting_time"
29 echo -e "\033[1;35mAverage Turnaround Time: \033[0
    m$average_turnaround_time"
30 echo -e "\033[1;36m
    =====\033[0m
    "
31 }
32
33 # Read input from user
34 echo -e "\033[1;36mEnter the number of processes:\033[0m"
35 read n
36
37 # Declare arrays for arrival time, burst time, completion time
38 declare -a arrival_times
39 declare -a burst_times
40 declare -a remaining_times
41 declare -a completion_times
42
43 # Read process details
44 for ((i=0; i<n; i++)); do
45     echo -e "\033[1;32mEnter details for Process $((i+1)):\033[0m"
46     echo -e "\033[1;36mEnter Arrival Time for Process $((i+1))
        :\033[0m"
47     read arrival_times[$i]
48     echo -e "\033[1;36mEnter Burst Time for Process $((i+1)):\033[0m
        "
49     read burst_times[$i]
50     remaining_times[$i]=${burst_times[$i]} # Initially, remaining
        time equals burst time
51     completion_times[$i]=0
52 done
53
54 echo -e "\033[1;36mEnter Time Quantum:\033[0m"
55 read time_quantum
56
57 # Initializing variables
58 current_time=0
59 processes_completed=0
60
61 # Array to keep track of process completion status
62 is_completed=()
63 for ((i=0; i<n; i++)); do
64     is_completed[$i]=0 # Initially, no processes are completed
65 done
66
67 # Round Robin algorithm logic
68 echo -e "\n\033[1;34mStarting Round Robin Scheduling Algorithm
    ...\033[0m"
69
70 # Loop over processes until all are completed
71 while [ $processes_completed -lt $n ]; do

```

```

72  executed=false
73
74  # Go through each process and attempt to execute it
75  for ((i=0; i<n; i++)); do
76      if [ ${arrival_times[$i]} -le $current_time ] && [ ${
          remaining_times[$i]} -gt 0 ] && [ ${is_completed[$i]} -eq 0
          ]; then
77          # Check if a process has remaining burst time
78          if [ ${remaining_times[$i]} -le $time_quantum ]; then
79              current_time=$((current_time + remaining_times[$i]))
80              completion_times[$i]=$current_time
81              remaining_times[$i]=0 # Process completed
82
83              is_completed[$i]=1 # Mark as completed
84              echo -e "\033[1;33mProcess P$((i+1)) completed at time
                  $current_time.\033[0m"
85          else
86              current_time=$((current_time + time_quantum))
87              remaining_times[$i]=$((remaining_times[$i] - time_quantum)
              )
88              echo -e "\033[1;33mProcess P$((i+1)) executed for
                  $time_quantum units, current time: $current_time.\033[0
                  m"
89          fi
90          executed=true
91      fi
92  done
93
94  # If no process was executed (idle time), move the current time
      forward
95  if [ "$executed" = false ]; then
96      current_time=$((current_time + 1))
97  fi
98
99  # Count how many processes have completed
100  processes_completed=0
101  for ((i=0; i<n; i++)); do
102      if [ ${is_completed[$i]} -eq 1 ]; then
103          processes_completed=$((processes_completed + 1))
104      fi
105  done
106 done
107
108 # Calculate times and print results
109 calculate_times

```

Listing 2.5: Round Robin Scheduling Script

2.5.6 Priority Scheduling Algorithm

```
1  #!/bin/bash
2
3  # Function to calculate waiting time and turnaround time
4  calculate_wait_turnaround() {
5      local -n arrival=$1
6      local -n burst=$2
7      local -n priority=$3
8      local -n completion=$4
9      local -n waiting=$5
10     local -n turnaround=$6
11     local n=${#arrival[@]}
12
13     # Calculate waiting and turnaround times
14     for (( i=0; i<n; i++ )); do
15         # Turnaround Time = Completion Time - Arrival Time
16         turnaround[$i]=$((completion[i] - arrival[i]))
17
18         # Waiting Time = Turnaround Time - Burst Time
19         waiting[$i]=$((turnaround[i] - burst[i]))
20     done
21 }
22
23 # Function for Non-Preemptive Priority Scheduling
24 non_preemptive_priority() {
25     local n=${#arrival[@]}
26     local -a waiting
27     local -a turnaround
28     local -a completion
29     local -a remaining_burst
30     local -a start_time
31     local min_priority min_priority_process min_burst
32
33     # Initialize variables
34     remaining_burst=("${burst[@]}")
35     completion=()
36     start_time=()
37
38     # Calculate completion time based on Non-Preemptive Priority
39     local current_time=0
40     for ((i = 0; i < n; i++)); do
41         min_priority=-1
42         min_priority_process=-1
43         min_burst=999999
44
45         # Find the process with the highest priority
46         for ((j = 0; j < n; j++)); do
47             if [[ "${arrival[$j]}" -le "$current_time" && "${remaining_burst[$j]}" -gt 0 && "${priority[$j]}" -lt "$min_priority" ]]; then
48                 min_priority="${priority[$j]}"
49                 min_priority_process=$j
50                 min_burst="${remaining_burst[$j]}"
51             fi
52         done
53
```



```

54      # Update current time and execution
55      if [[ $min_priority_process -ne -1 ]]; then
56          start_time[$min_priority_process]="$current_time"
57          completion[$min_priority_process]="${(start_time[
                    $min_priority_process] + remaining_burst[
                    $min_priority_process])}"
58          current_time="${completion[$min_priority_process]}"
59          remaining_burst[$min_priority_process]=0
60      fi
61  done
62
63  # Calculate Waiting and Turnaround Times
64  calculate_wait_turnaround arrival burst priority completion
        waiting turnaround
65
66  # Display Results
67  echo -e "\n\033[1;32m
        =====\033[0m"
68  echo -e "\033[1;32mProcess ID      Arrival Time      Burst Time
        Priority      Completion Time      Waiting Time
        Turnaround Time\033[0m"
69  echo -e "\033[1;32m
        =====\033[0m"
        "
70  for ((i=0; i<n; i++)); do
71      echo -e "\033[1;36m P${(i+1)}           ${arrival[i]}
                    ${burst[i]}           ${priority[i]}
                    ${completion[i]}       ${waiting[i]}
                    ${turnaround[i]}\033[0m"
72  done
73  echo -e "\033[1;32m
        =====\033[0m"
74  avg_wait=0
75  avg_turnaround=0
76  for ((i=0; i<n; i++)); do
77      avg_wait=$((avg_wait + waiting[i]))
78      avg_turnaround=$((avg_turnaround + turnaround[i]))
79  done
80  echo -e "\033[1;33mAverage Waiting Time: $((avg_wait / n))
        \033[0m"
81  echo -e "\033[1;33mAverage Turnaround Time: $((avg_turnaround
        / n))\033[0m"
82  echo -e "\033[1;32m
        =====\033[0m"
        "
83  }
84
85  # Function for Preemptive Priority Scheduling
86  preemptive_priority() {
87      local n=${#arrival[@]}
88      local -a waiting
89      local -a turnaround
90      local -a completion
91      local -a remaining_burst
92      local -a start_time
93
94      # Initialize variables
95      remaining_burst=("${burst[@]}")

```

```

96 completion=()
97 start_time=()
98
99 # Calculate completion time based on Preemptive Priority
100 local current_time=0
101 local completed=0
102 while [[ $completed -lt $n ]]; do
103     min_priority=-1
104     min_priority_process=-1
105     min_burst=999999
106
107     for ((j = 0; j < n; j++)); do
108         if [[ "${arrival[$j]}" -le "$current_time" && "${remaining_burst[$j]}" -gt 0 && "${priority[$j]}" -lt "$min_priority" ]]; then
109             min_priority="${priority[$j]}"
110             min_priority_process=j
111             min_burst="${remaining_burst[$j]}"
112         fi
113     done
114
115     if [[ $min_priority_process -ne -1 ]]; then
116         if [[ -z "${start_time[$min_priority_process]}" ]]; then
117             start_time[$min_priority_process]="$current_time"
118         fi
119
120         remaining_burst[$min_priority_process]=$((remaining_burst[$min_priority_process] - 1))
121         current_time=$((current_time + 1))
122
123         if [[ ${remaining_burst[$min_priority_process]} -eq 0 ]]; then
124             completion[$min_priority_process]="$current_time"
125             completed=$((completed + 1))
126         fi
127     else
128         current_time=$((current_time + 1))
129     fi
130 done
131
132 # Calculate Waiting and Turnaround Times
133 calculate_wait_turnaround arrival burst priority completion
134     waiting turnaround
135
136 # Display Results
137 echo -e "\n\033[1;32m
138     =====\033[0m"
139 echo -e "\033[1;32mProcess ID      Arrival Time      Burst Time
140     Priority      Completion Time      Waiting Time
141     Turnaround Time\033[0m"
142 echo -e "\033[1;32m
143     =====\033[0m"
144
145 for ((i=0; i<n; i++)); do
146     echo -e "\033[1;36m P$((i+1))           ${arrival[i]}
147             ${burst[i]}           ${priority[i]}
148             ${completion[i]}           ${waiting[i]}

```

```

141         done
142         echo -e "\033[1;32m
           =====\033[0m"
           "
143         avg_wait=0
144         avg_turnaround=0
145         for ((i=0; i<n; i++)); do
146             avg_wait=$((avg_wait + waiting[i]))
147             avg_turnaround=$((avg_turnaround + turnaround[i]))
148         done
149         echo -e "\033[1;33mAverage Waiting Time: $((avg_wait / n))
           \033[0m"
150         echo -e "\033[1;33mAverage Turnaround Time: $((avg_turnaround
           / n))\033[0m"
151         echo -e "\033[1;32m
           =====\033[0m"
152     }
153
154     # Taking user input for arrival, burst, and priority times for the
       processes
155     echo -e "\n\033[1;34m
           =====\033[0m"
156     echo -e "\033[1;34m Enter the number of processes:\033[0m"
157     read n
158
159     # Create empty arrays for arrival, burst, and priority times
160     arrival=()
161     burst=()
162     priority=()
163
164     # Read process details from the user
165     for ((i=0; i<n; i++)); do
166         echo -e "\n\033[1;34m Enter Arrival Time for Process P$((i+1))
           : \033[0m"
167         read arrival_time
168         arrival+=($arrival_time)
169
170         echo -e "\033[1;34m Enter Burst Time for Process P$((i+1)):
           \033[0m"
171         read burst_time
172         burst+=($burst_time)
173
174         echo -e "\033[1;34m Enter Priority for Process P$((i+1)):
           \033[0m"
175         read priority_time
176         priority+=($priority_time)
177     done
178
179     # Asking user to select the scheduling method
180     echo -e "\n\033[1;35mSelect scheduling type:\033[0m"
181     echo -e "\033[1;35m1. Non-Preemptive\033[0m"
182     echo -e "\033[1;35m2. Preemptive\033[0m"
183     echo -e "\033[1;35mEnter your choice:\033[0m"
184     read scheduling_type
185
186     # Running the selected scheduling algorithm
187     case $scheduling_type in

```

```
188     1)
189         non_preemptive_priority
190         ;;
191     2)
192         preemptive_priority
193         ;;
194     *)
195         echo -e "\033[1;31mInvalid choice! Please select either 1
196             (Non-Preemptive) or 2 (Preemptive).\033[0m"
197     esac
```

Chapter 3

Performance Evaluation

3.1 Introduction

The performance evaluation of the **CPU Scheduler** is a critical aspect of this project, as it determines the efficiency and practicality of the implemented algorithms in diverse scenarios. This chapter presents the methodology, simulation environment, and analysis of results obtained from testing the scheduling algorithms. The evaluation focuses on key metrics such as waiting time, turnaround time, and CPU utilization to compare the performance of the five algorithms.

3.2 Simulation Environment

3.2.1 System Configuration

The simulation was conducted in a controlled environment using the following system specifications:

- **Operating System:** FydeOS
- **Processor:** Intel Core i5
- **RAM:** 8 GB
- **Shell:** Bash (Bourne Again Shell)
- **Utilities:** awk, bc, and logging tools for results analysis

3.2.2 Input Parameters

The input parameters for the simulation include:

- **Number of Processes (n):** Varies from 5 to 50.
- **Arrival Time:** Randomly generated or user-defined.

- **Burst Time:** Randomly generated or user-defined.
- **Priority:** Randomly generated or user-defined (used in Priority Scheduling).
- **Time Quantum:** Fixed or user-defined for Round Robin scheduling.

3.3 Performance Metrics

The following metrics were used to evaluate the performance of the algorithms:

- **Completion Time:** Time at which a process completes execution.
- **Waiting Time:** Time a process spends waiting in the ready queue.
- **Turnaround Time:** Total time from arrival to completion of a process.
- **Average Waiting Time:** Average of waiting times for all processes.
- **Average Turnaround Time:** Average of turnaround times for all processes.
- **CPU Utilization:** Percentage of CPU time utilized for executing processes.

3.4 Results and Analysis

3.4.1 First-Come-First-Serve (FCFS)

- **Observations:** FCFS exhibits simplicity but suffers from the "convoy effect," where long processes delay shorter ones.
- **Performance:** Provides fair scheduling but results in high average waiting and turnaround times when process arrival times vary significantly.
- **Use Case:** Best suited for batch processing systems.

3.4.2 Shortest Job First (SJF)

- **Observations:** SJF minimizes average waiting and turnaround times by prioritizing processes with shorter burst times.
- **Performance:** Performs optimally when burst times are predictable but is prone to starvation for longer processes.
- **Use Case:** Suitable for environments with predictable process execution times.

3.4.3 Shortest Remaining Time First (SRTF)

- **Observations:** A preemptive variant of SJF, SRTF provides better responsiveness but incurs higher context-switching overhead.
- **Performance:** Balances responsiveness and efficiency but increases CPU overhead due to frequent preemptions.
- **Use Case:** Ideal for real-time systems requiring high responsiveness.

3.4.4 Round Robin (RR)

- **Observations:** RR ensures fairness and avoids starvation by allocating a fixed time quantum to each process.
- **Performance:** Average waiting and turnaround times depend heavily on the chosen time quantum. Too small a quantum increases context switching; too large reduces responsiveness.
- **Use Case:** Best suited for time-sharing systems.

3.4.5 Priority Scheduling

- **Observations:** Prioritizes processes based on their priority levels, with options for preemptive and non-preemptive execution.
- **Performance:** Performance varies depending on the priority distribution. Prone to starvation if low-priority processes are repeatedly bypassed.
- **Use Case:** Effective in systems where certain processes require higher priority, such as real-time or embedded systems.

3.5 Output

3.5.1 Output

The outputs are given below-

```
#####
#           CPU Scheduling Simulation Program           #
#####

Please select a scheduling algorithm to run:

[1] First-Come-First-Serve (FCFS)
[2] Shortest Job First (SJF)
[3] Shortest Remaining Time First (SRTF)
[4] Round Robin (RR)
[5] Priority Scheduling (Preemptive/Non-Preemptive)
[Q] Exit

Enter your choice:
```

Figure 3.1: Main Interface

```
##### FCFS Scheduling #####
Executes processes in the order of their arrival time.

Enter the number of processes:
3
Enter Arrival Time for Process 1:
0
Enter Burst Time for Process 1:
4
Enter Arrival Time for Process 2:

Invalid input! Enter a non-negative integer for arrival time.
4
Enter Burst Time for Process 2:

Invalid input! Enter a positive integer for burst time.
6
Enter Arrival Time for Process 3:
5
Enter Burst Time for Process 3:
8

Results:
-----
Process ID    Arrival Time    Burst Time    Completion Time    Waiting Time    Turnaround Time
P1            0              4             4                  0              4
P2            4              6             10                 -10            -4
P3            5              8             13                 -13            -5
-----
Average Waiting Time: -7.67
Average Turnaround Time: -1.67

Press any key to return to the main menu...
```

Figure 3.2: FCFS Scheduling

SJF Scheduling Results:

Process ID	Arrival Time	Burst Time	Completion Time	Waiting Time	Turnaround Time
P1	0	4	4	0	4
P2	6	5	9	-2	3
P3	8	6	15	1	7

Average Waiting Time: 0
Average Turnaround Time: 3

Figure 3.3: SJF Scheduling

Starting SRTF Scheduling Algorithm...

Process P1 completed at time 5.
Process P2 completed at time 10.
Process P3 completed at time 16.

Process ID	Arrival Time	Burst Time	Completion Time	Waiting Time	Turnaround Time
P1	0	5	5	5-0-5	5-0
P1	6	4	10	10-6-4	10-6
P1	8	6	16	16-8-6	16-8

Average Waiting Time: .6666666666666666
Average Turnaround Time: 5.6666666666666666

Press any key to return to the main menu...

Figure 3.4: SRTF Scheduling

ROUND ROBIN SCHEDULING RESULTS

Process ID	Arrival Time	Burst Time	Completion Time	Waiting Time	Turnaround Time
P1	0	4	4	4-0-4	4-0
P2	5	6	13	13-5-6	13-5
P3	8	6	17	17-8-6	17-8

Average Waiting Time: 1.6666666666666666
Average Turnaround Time: 7.0000000000000000

Press any key to return to the main menu...

Figure 3.5: Round Robin Scheduling

```

Running Priority Scheduling...
=====
Enter the number of processes:
2
Enter Arrival Time for Process P1:
3
Enter Burst Time for Process P1:
4
Enter Priority for Process P1:
6
Enter Arrival Time for Process P2:
4
Enter Burst Time for Process P2:
8
Enter Priority for Process P2:
6
Select scheduling type:
1. Non-Preemptive
2. Preemptive
Enter your choice:

```

Figure 3.6: Priority Scheduling

3.6 Comparison of Algorithms

Table 3.1: Performance Comparison of Scheduling Algorithms

Metric	FCFS	SJF	SRTF	RR	Priority
Average Waiting Time	High	Low	Moderate	Moderate	Variable
Average Turnaround Time	High	Low	Moderate	Moderate	Variable
CPU Utilization	High	High	Moderate	Moderate	High
Context-Switching Overhead	Low	Low	High	Moderate	Variable
Starvation Risk	Low	High	Moderate	Low	High
Best Use Case	Batch Systems	Predictable	Real-Time	Time-Sharing	Real-Time/Embedded

Chapter 4

Conclusion and Future Work

4.1 Discussion and Summary

The **CPU Scheduler** project successfully implements and evaluates five widely used CPU scheduling algorithms: First-Come-First-Serve (FCFS), Shortest Job First (SJF), Shortest Remaining Time First (SRTF), Round Robin (RR), and Priority Scheduling (Preemptive and Non-Preemptive). Through the design and implementation process, the project highlights the importance of CPU scheduling in process management within operating systems.

The simulator provides:

- A modular and interactive platform for testing and analyzing CPU scheduling algorithms.
- Accurate calculation of critical metrics such as waiting time, turnaround time, and completion time.
- A comprehensive logging system for detailed analysis and debugging.
- User-friendly interfaces with configurable settings for time quantum, priority levels, and more.

4.2 Key Insights

From the performance evaluation in Chapter 3, several key insights were observed:

- No single algorithm is optimal for all scenarios; the choice depends on system requirements such as fairness, responsiveness, and efficiency.
- Algorithms like SJF and SRTF minimize average waiting and turnaround times but may cause starvation.
- RR provides fairness and prevents starvation, making it suitable for time-sharing systems, though the time quantum must be carefully chosen to balance responsiveness and overhead.

- **Priority Scheduling**, while effective in real-time and critical systems, can lead to starvation of lower-priority processes if not managed properly.

4.3 Limitations

Despite the success of the project, some limitations were identified:

- **Dynamic Process Input:** The simulator requires predefined input parameters. Real-world scheduling often involves dynamic process arrivals, which this simulator handles only to a limited extent.
- **Scalability:** While designed to handle up to 50 processes, the computational overhead increases significantly for higher process counts, especially in preemptive algorithms like SRTF.
- **Hardware Simulation:** The project does not simulate hardware-level constraints, such as I/O operations and cache performance, which could affect real-world scheduling.

4.4 Scope of Future Work

This project serves as a foundational framework for CPU scheduling simulations. Several enhancements and extensions can improve its functionality and applicability:

- **Dynamic Process Management:** Extend the simulator to handle processes arriving dynamically during execution.
- **Additional Algorithms:** Implement and analyze other scheduling algorithms, such as Multi-level Queue Scheduling or Multi-level Feedback Queue Scheduling.
- **Graphical User Interface (GUI):** Develop a GUI for better visualization of scheduling processes and metrics.
- **Hardware Simulation:** Incorporate hardware-level constraints to simulate real-world CPU behavior.
- **Machine Learning Integration:** Use machine learning models to predict optimal scheduling strategies based on historical data.
- **Performance Optimization:** Optimize the simulator for scalability to handle a larger number of processes efficiently.

4.5 Concluding Remarks

The **CPU Scheduler** project demonstrates the practical application of operating system principles in process management. By simulating key CPU scheduling algorithms, it bridges the gap between theoretical understanding and practical implementation, providing users with an insightful tool for analysis and education.

This project serves as a stepping stone for further research and development in CPU scheduling, with potential applications in educational environments, real-time systems, and advanced operating system design. With the proposed enhancements, the **CPU Scheduler** can evolve into a powerful and versatile tool for both academia and industry.

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